H

RAFFLES INSTITUTION

2013 Year 6 Preliminary Examination
Higher 2

CANDIDATE NAME	

CIVICS GROUP	1	3	S	0	3		INDEX NUMBER						

BIOLOGY

Paper 2 Core paper

9648/02
16th SEPTEMBER 2013
2 hours

Additional materials: Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **either ONE** question.

At the end of the examination, **hand in your essay SEPARATELY.**
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/12
2	/13
3	/12
4	/13
5	/11
6	/10
7	/9
Section B	
8 or 9	/20
Total	/100

This document consists of **19** printed pages.

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Section A

Answer **all** the questions in this section.

1 **Fig. 1.1** shows the sequence of bases in a section of a gene coding for the first seven amino acids of a polypeptide chain.

A A A C T G T T T C T G C T G <u>G</u> A A C A T

Fig. 1.1

(a) **(i)** Methionine is coded by the initiation codon. On **Fig. 1.1**, label the 3' and 5' end of the DNA template strand. [1]

(ii) List the corresponding codons in the mRNA transcribed from the template in **Fig. 1.1**.

.... [1]

The polypeptide was hydrolysed to release the first seven amino acids. It contained four different amino acids. The number of each type obtained is shown in the **Table 1.1**.

Amino acid	Number present
Phe	2
Met	1
Lys	1
Gln	3

Table 1.1

(iii) Using information in **Table 1.1**, work out the order of amino acids in the polypeptide.

.... [1]

(b) **(i)** Explain why a single base deletion in the DNA sequence is more likely to produce non-functional proteins than a single base substitution.

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.... [3]

(ii) Describe the effect of replacing guanine (underlined in **Fig.1.1**) with adenine.

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(c) Explain how the structure of the nuclear envelope facilitates translation.

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(d) A variety of related proteins can be produced from a single gene. State the process and explain how the same gene can produce different protein products.

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[Total : 12]

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2 Chromosome aberrations can have devastating effects on human health.

(a) Down syndrome is a disorder caused by an aberration in chromosomal number. Individuals with Down syndrome have three copies of chromosome 21, which typically results in some delay in cognitive ability and physical growth, as well as a characteristic set of facial features.

Fig. 2.1 is a diagrammatic representation of chromosome 21, showing the location of two different RFLP loci.

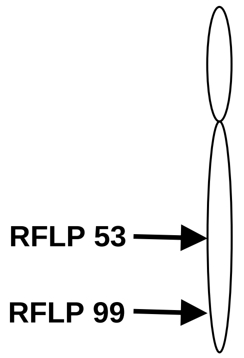

Fig. 2.1

Fig. 2.2 is a pedigree of a family and the results following restriction digestion, gel electrophoresis and Southern blotting with specific probes of the genetic material of some family members.

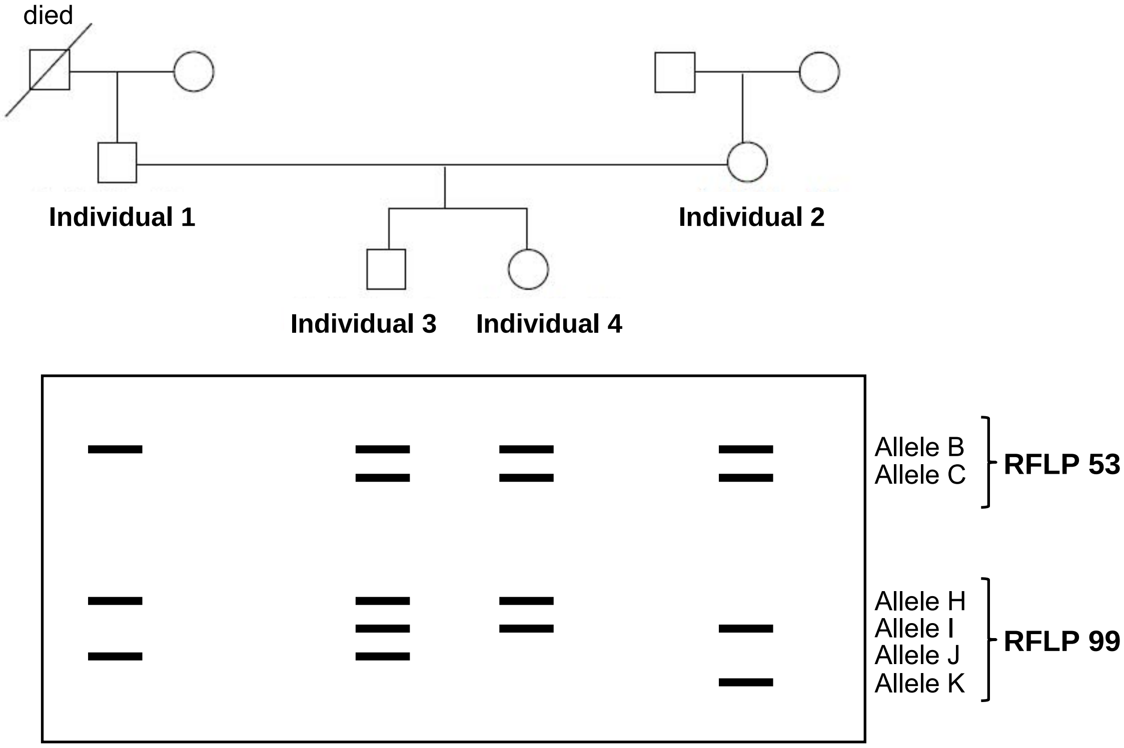

Fig. 2.2

(i) Identify which individual (**3** or **4**) has three copies of chromosome 21. Give a reason for your answer.

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……. [2]

(ii) Non-disjunction of chromosomes during the formation of gametes would have resulted in gametes with two copies of chromosome 21.
In which parent (individual **1** or **2**) did the non-disjunction of chromosomes occur?

……. [1]

(iii) State when the non-disjunction event occurred. Explain your answer.

………

………

………

……. [2]

(iv) Draw chromosome(s) 21 in the gamete that resulted in the individual having Down syndrome. Clearly label the alleles at the RFLP loci on each homologue of chromosome 21.

gamete

[1]

For Examiner's Use

(b) **Fig. 2.3** shows the formation of a Philadelphia chromosome which is a chromosomal abnormality that results from a reciprocal translocation between chromosome 9 and 22. This chromosomal abnormality is associated with chronic myelogenous leukemia which is considered a form of cancer.

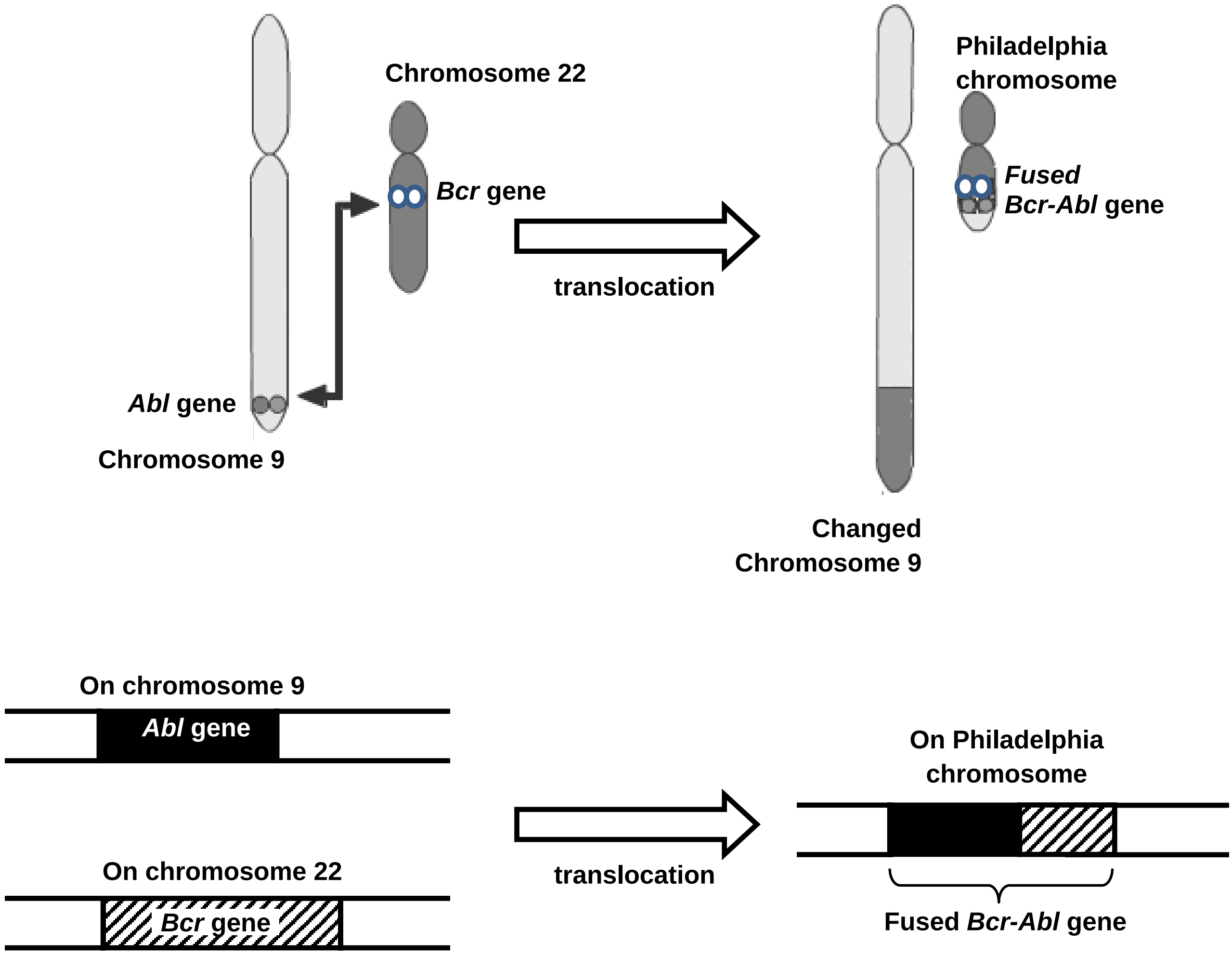

Fig. 2.3

The A*bl* gene normally codes for a receptor involved in signal transduction that results in the cell proliferation.
Using the information in **Fig. 2.3**, suggest how the fused gene product can result in cancer.

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.... [3]

(c) Cancer cells require tetrahydrofolate to make purines and pyrimidines for cell proliferation. Conversion of precursors to tetrahydrofolate is carried out by the enzyme dihydrofolate reductase (DHFR). Competitive inhibition of DHFR by administering the drug Methotrexate (MTX) is a key treatment for certain cancers such as acute lymphoblastic leukemia.

(i) Using the information provided, comment on the structure of Methotrexate.

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.... [1]

(ii) However, researchers have since found that one of the main causes of failure in the treatment of acute lymphoblastic leukemia is MTX resistance due to the amplification of the DHFR gene.
During the early stages of the research, scientists suggested a mechanism for the amplification of the DHFR gene as shown in **Fig. 2.4**.

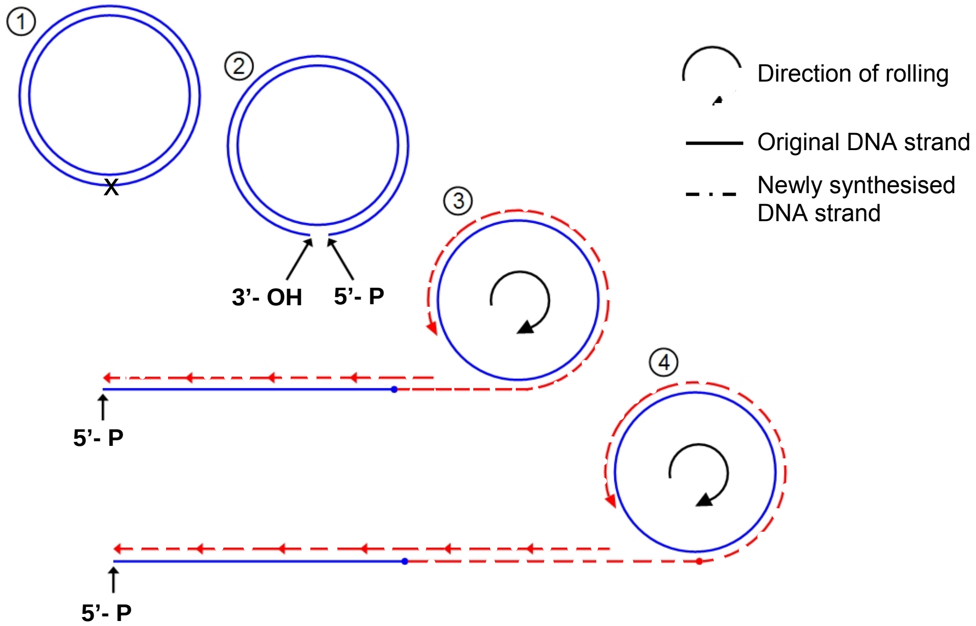

Fig. 2.4

With reference to **Fig. 2.4**, describe the process.

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.... [3]

[Total : 13]

For Examiner's Use

3(a) A study was conducted on the *lac* operon of *E. coli*. **Fig. 3.1** below shows the activity of β-galactosidase as a function of time for a culture of *E. coli*. The bacterial cells were initially grown in glycerol as the only carbon source. At time A, lactose was added to the culture. When the lactose supply is exhausted at time B, glucose was added.

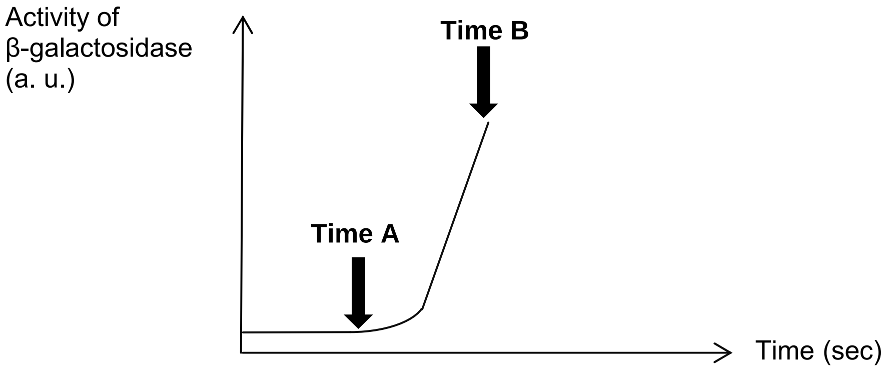

Fig. 3.1

(i) Why is the *lac* operon considered inducible?

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…………………………………………………………………………………………………. [2]

(ii) Explain the change in β-galactosidase activity from time **A** to **B**. [4]

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(iii) Draw in **Fig. 3.1** what a continuation of the graph would look like from time B.

[1]

For Examiner's Use

(b) **Fig. 3.2** shows the arrangement of the *lac* operon and its regulatory gene.

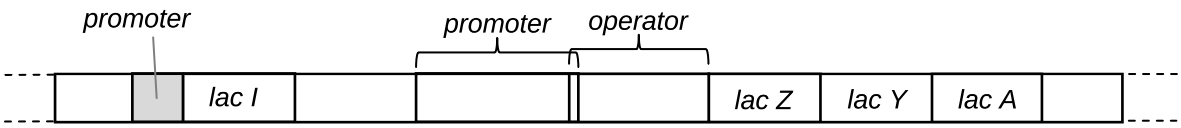

Fig. 3.2

A series of mutations was introduced to the *lac* operon.

(i) The operator is deleted. Assuming lactose is present, what is the effect of this on the transcription of *lac Z, lac Y* and *lac A* genes?

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…………………………………………………………………………………………….. [2]

(ii) A point mutation was introduced into *lac Z*, resulting in the production of a truncated non-functional protein. Assuming lactose is present, explain if the proteins encoded by *lac Y* and *lac A* genes are produced?

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(c) An *E. coli* strain was isolated from the intestines of a mammal. The bacterium was found to have two different copies of the *lac* operon. Describe a genetic transfer process that resulted in this observation.

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…………………………………………………………………………………………….. [1]

[Total : 12]

4 An **absorption** spectrum is a graph of the absorption of different wavelengths of light by a photosynthetic pigment.
An **action** spectrum is a graph of the rate of photosynthesis at different wavelengths of light. **Fig. 4.1** shows the absorption spectra of chlorophyll a and chlorophyll b as well as an action spectrum.

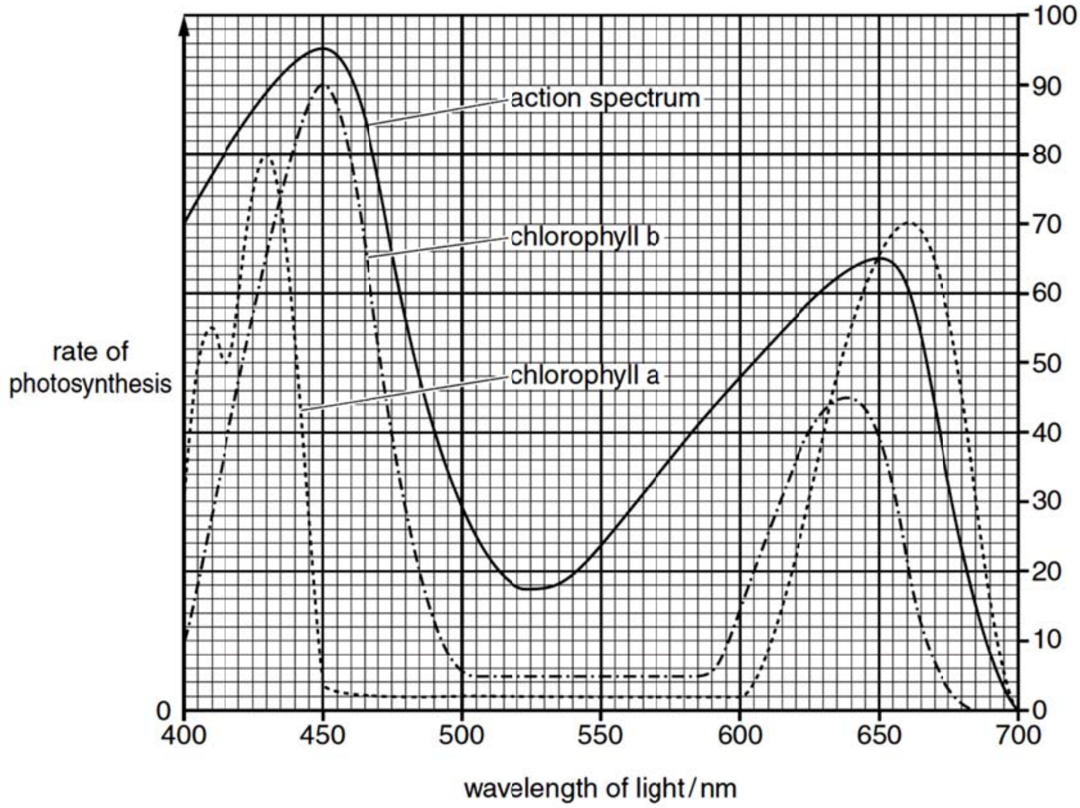

Fig. 4.1

(a) With reference to **Fig. 4.1**,

(i) compare the **absorption** spectra of chlorophyll a and chlorophyll b.

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.... [2]

(ii) explain the shape of the **action** spectrum.

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.... [2]

For Examiner's Use

(b) The light-dependent stage of photosynthesis takes place on the thylakoids of the chloroplast. **Fig. 4.2** represents a section through a thylakoid to show some of the components involved in the light-dependent stage.

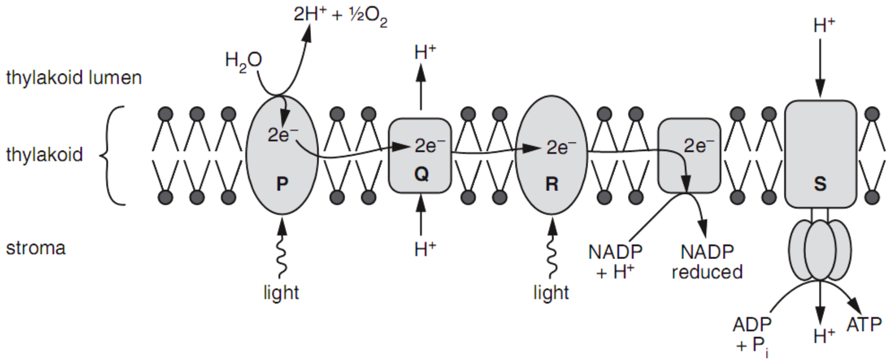

Fig. 4.2

(i) Identify the structures labelled R and S.

R : .... **S** : ....[2]

(ii) Atrazine is a widely used herbicide (weed killer). It binds to a chloroplast protein involved in electron transfer between photosystems.
Explain how atrazine works as an effective herbicide.

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.... [3]

For Examiner's Use

(c) In an investigation, mammalian liver cells were homogenised (broken up) and the resulting homogenate centrifuged. Samples of the complete homogenate and samples with only mitochondria were incubated with:

1 glucose
2 pyruvate
3 glucose and cyanide
4 pyruvate and cyanide

Cyanide inhibits oxidative phosphorylation.

After incubation, the presence or absence of carbon dioxide and lactate in each sample was determined. The results are summarised in **Table 4.1**.

	Sample of homogenate			
	Complete		Only mitochondria	
	Carbon dioxide	Lactate	Carbon dioxide	Lactate
1 glucose	√	√	X	X
2 pyruvate	√	√	√	X
3 glucose and cyanide	X	√	X	X
4 pyruvate and cyanide	X	√	X	X

***X* = absent √ = present**

Table 4.1

(i) Explain why carbon dioxide is produced when mitochondria are incubated with pyruvate but **not** when they are incubated with glucose.

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.... [2]

(ii) With reference to **Table 4.1,** explain the difference in the results when the complete cell homogenate is incubated with **2** and **4**.

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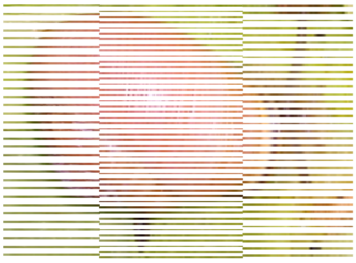
.... [2]

[Total : 13]

5 Grove snails are known to have brightly coloured shells, with the alleles C^P and C^Y coding for pink and yellow coloured shells respectively. Shells of the grove snails have different banding patterns as well, with allele B^B conferring a shell with bands and allele B^O conferring an unbanded shell.



Banded snail Unbanded snail

A grove snail with a banded pink shell is crossed with another grove snail with an unbanded yellow shell, and the resultant F_1 generation all had unbanded pink shells. The F_1 were test-crossed and gave rise to 250 offspring with the following phenotypes:

banded pink shell	121
unbanded pink shell	14
banded yellow shell	11
unbanded yellow shell	104

(a) State the expected offspring phenotypic ratio of the test cross.

.... [1]

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

χ^2 test: $\chi^2 = \Sigma \frac{(O - E)^2}{E}$ $\nu = c - 1$

Key to symbols:
Σ = sum of …
ν = degrees of freedom
c = number of classes
O = observed value
E = expected value

(b) **(i)** Showing your working clearly, calculate the χ^2 value in the space below.

For Examiner's Use

[2]

(ii) Using the χ^2 table, explain the conclusion drawn from the calculated χ^2 value in b(i).

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……………………………………………………………………………………………….. [2]

(c) With the use of a genetic diagram, explain the observed results of the test cross.

[5]

The same test cross was carried out another time and the resulting number of banded pink snails and unbanded yellow snails were equal. The resulting number of unbanded pink snails was also equal to that of the banded yellow snails. Through this test cross, the 2 gene loci are determined to be 28 map units apart.

(d) State the observed phenotypic ratio of the offspring from this test cross.

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[Total : 11]

6 Homeostatic mechanisms maintain a constant environment in the body.
Fig. 6.1 shows changes in plasma glucose concentration that occurred in a person who went without food for some time.

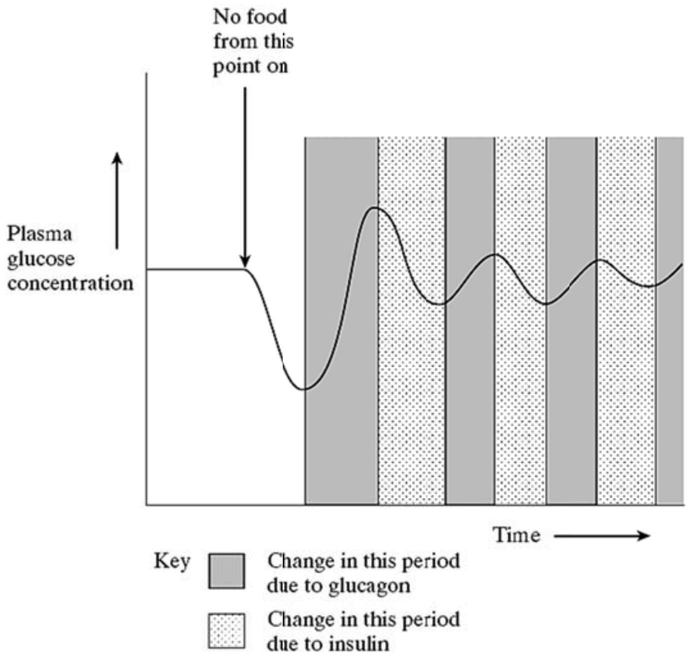

Fig. 6.1

(a) Use evidence from **Fig. 6.1** to explain the change in plasma glucose concentration.

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For Examiner's Use

.... [4]

Adrenaline is a signal molecule that stimulates similar cellular responses from liver cells and adipose cells in a way similar to that of glucagon.

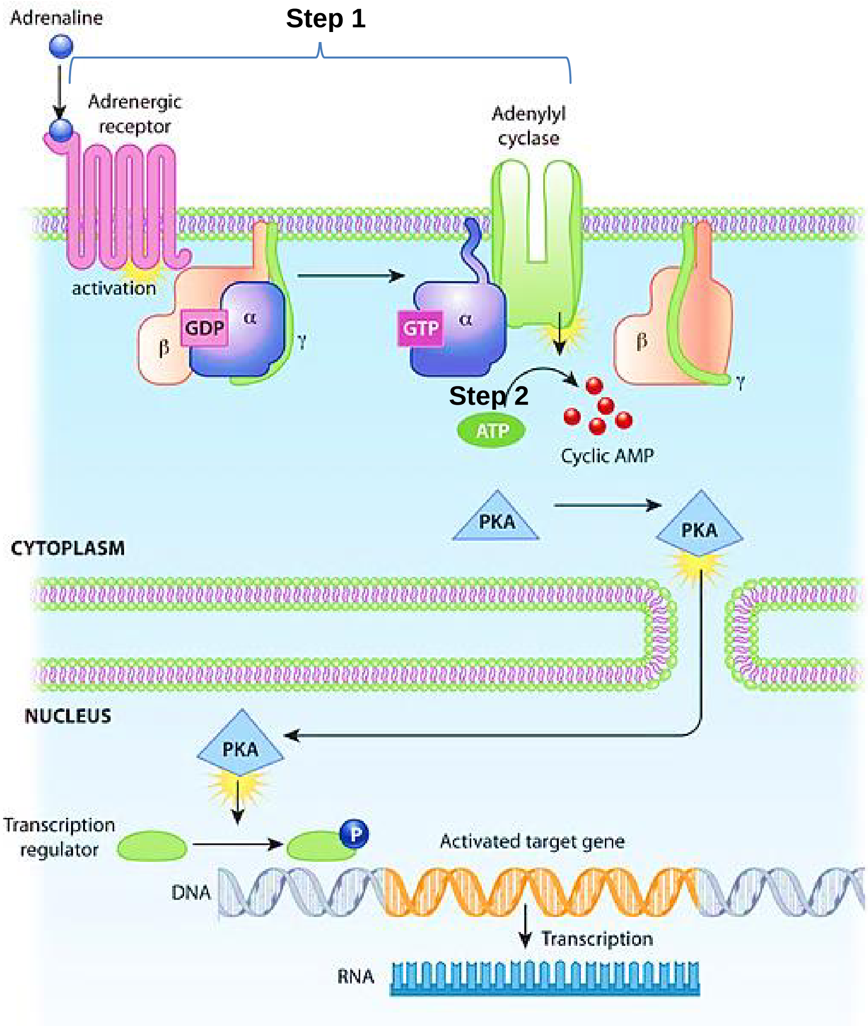

Fig. 6.2

(b) With reference to **Fig. 6.2**,

(i) describe the events occurring in step 1 after the binding of adrenaline molecule to the adrenergic receptor.

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.... [2]

(ii) explain the significance of step 2 that results in effective signal transduction.

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....

For Examiner's Use

…………………………………………………………………………………………………….. [2]

(iii) describe the series of events that occur after the activation of protein kinase A that results in a cellular response.

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[Total : 10]

7 For a long time, the prevailing idea among evolutionary biologists is that geographical barriers are a prerequisite for speciation. However, recently, more and more examples of an alternative speciation process which does not require a geographical barrier were discovered but this remains the exception rather than a norm.

(a) **(i)** What alternative speciation process are we referring to?

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One case study involved studying the palm genus *Howea* from Lord Howe Island, a tiny volcanic island in the Tasman Sea. The two related palm species found on the island have a different soil preference. The different types of soil are interspersed throughout the island and pollen from the palms is wind-dispersed.

For Examiner's Use

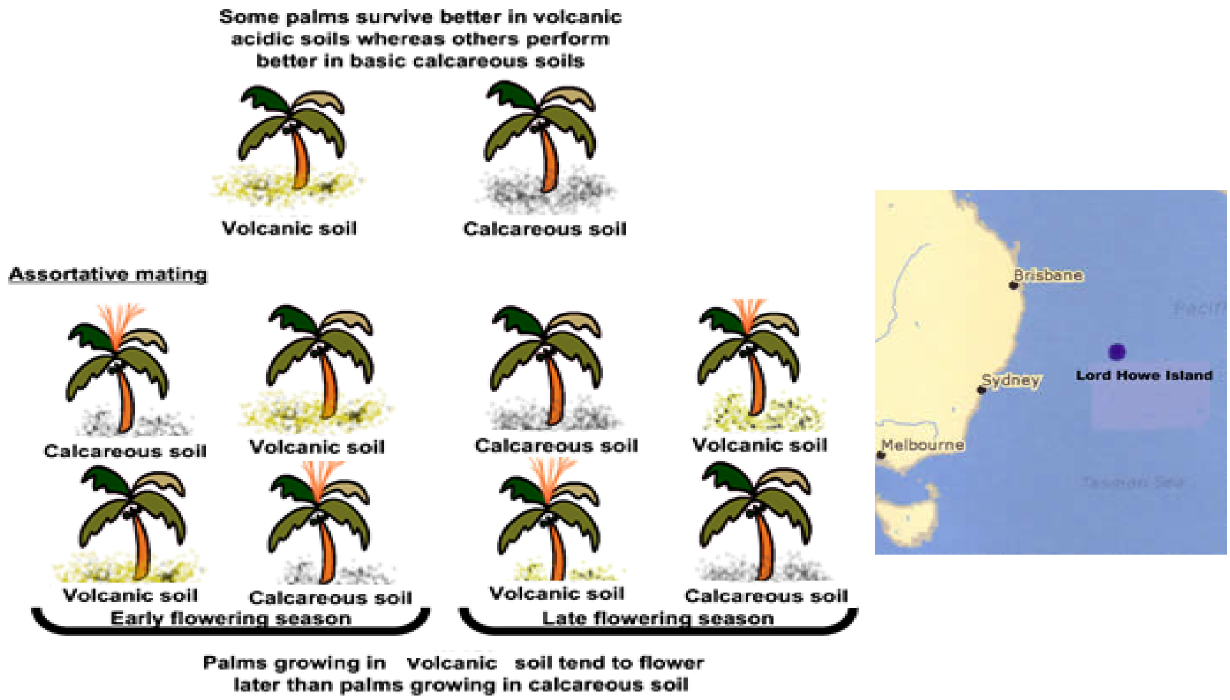

Fig. 7.1

(ii) Explain how speciation could have taken place to give rise to the two sister species from a common ancestor.

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.... [5]

(iii) Give an evidence why scientists can rule out geographical isolation mechanisms?

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.... [1]

For Examiner's Use

(b) *Howea belmoreana* goes by the names, curly palm, kentia palm and Belmore sentry palm while *Howea forsteriana* goes by the names thatch palm and kentia palm.

Explain why the scientific name is better than the common names.

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…….. [2]

[Total: 9]

Section B
Answer EITHER 8 OR 9.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8 **(a)** Explain how meiosis can contribute to genetic variation in offspring. [6]

(b) Discuss how bacteria evolve through natural selection and the implication in antibiotic resistance. [8]

(c) Describe the link between the frequency of sickle cell anaemia allele and the number of cases of malaria. [6]

[Total: 20]

9 **(a)** Explain how gene expression can be controlled at the translational level and describe the mechanisms present in prokaryotes and eukaryotes. [8]

(b) Explain how the structure of specific transcription factors is linked to their roles in control of eukaryotic gene expression. [7]

(c) With reference to changes in protein structure, explain how the level of protein activity can be regulated post-translationally. [5]

For Examiner's Use

[Total: 20]

---- End of Paper -----